Models Do Not Consistently Recognize or Reflect the Increased Difficulty in Hard Questions In Table 3, we analyze how response lengths vary across three subsets of hard questions: the Shared-Hard set $Q_{\cap}^{\text{hard}}$ (questions that are hard by both models), the $\mathcal{M}_2$-advantage set $Q_1^{\text{hard}} \setminus Q_2^{\text{hard}}$ (questions that are hard for $\mathcal{M}_1$ but of lower difficulty for $\mathcal{M}_2$), and the $\mathcal{M}_1$-advantage set $Q_2^{\text{hard}} \setminus Q_1^{\text{hard}}$. Following the logic from our analysis of easy questions, if a model can recognize increased difficulty, we would expect it to generate longer responses for questions in the Shared-Hard set compared to those in its own advantage set. However, as shown in Table 3, we do not observe a clear pattern. Instead, the average token lengths in the Shared-Hard subsets are sometimes shorter than those in each model's specific advantage set. This suggests that, when the questions are too challenging, models may struggle to be aware of, and respond to, increased difficulty in hard questions. The models may underestimate the difficulty of the problem and respond with unwarranted confidence, or lack the necessary knowledge and capacity to attempt a response whose length reflects the increased complexity of the question.

Length and Perplexity Pattern on Question Difficulty In Figure 3, we analyze how response length and perplexity vary on questions with different accuracy, which serves as a finer-grained proxy for question difficulty. We observe that average response length initially increases and then decreases as accuracy improves (i.e., as questions become easier), rather than decreasing monotonically—except for R1-Distill on GSM8K. When examining perplexity, we find that for R1-Preview on GSM8K, perplexity first increases as accuracy rises from 0 to 0.2, then decreases, indicating potential overconfidence on these most difficult questions. In contrast, for all other model-dataset pairs, perplexity decreases steadily as accuracy increases, suggesting that models become more confident as questions become easier. These trends in Figure 3 highlight two key failure modes when questions exceed a model's capability. In Appendix D, we show the pairwise heatmaps of accuracy, token length and perplexity. We observe that for questions with 0 accuracy, response lengths are more broadly distributed across a wide range of token lengths, rather than concentrated in the high token-length region. In contrast, for questions with higher accuracy, response lengths are more tightly clustered around shorter lengths.

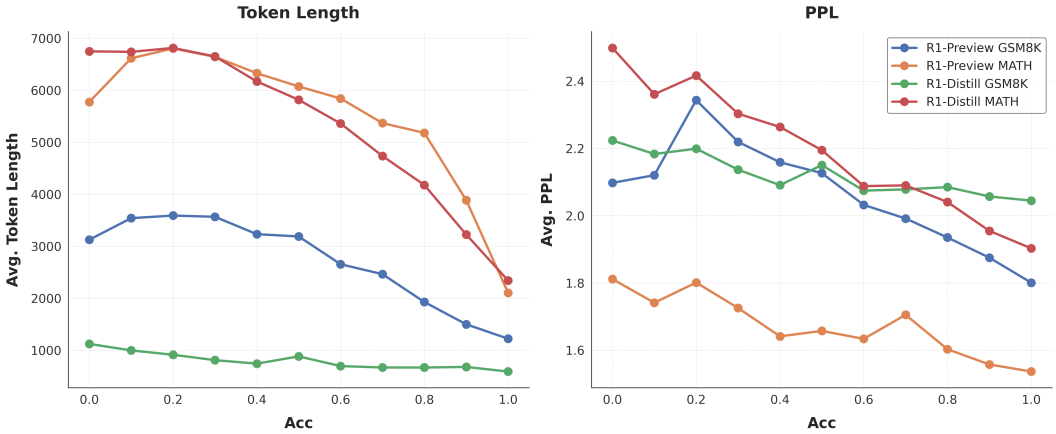

Figure 3: Average token length and perplexity of model responses across questions with different accuracy.

6 Effect of Length Preference Optimization

Prior work has explored length preference optimization as a strategy to balance correctness and brevity. A common approach involves generating multiple samples per input and constructing preference pairs that favor correct and/or shorter responses. For example, Yang et al. (2025) select the shortest correct response as the preferred one and use both the longest incorrect and shortest incorrect responses as negative samples. Similarly, Shen et al. (2025) propose a dual-pairing scheme: (1) preferring shorter responses among correct answers, and (2) penalizing shorter responses among incorrect ones. Although intuitive, these methods face two key practical limitations: (a) they require generating many samples per input (typically $N \geq 8$), which is often more computationally expensive than the

actual fine-tuning process, and the utility of generated samples are low–many samples are filtered out when constructing the preference pair; and (b) constructing these preference pairs requires data with ground-truth answers, limiting their applicability in settings where only unlabeled data is available.

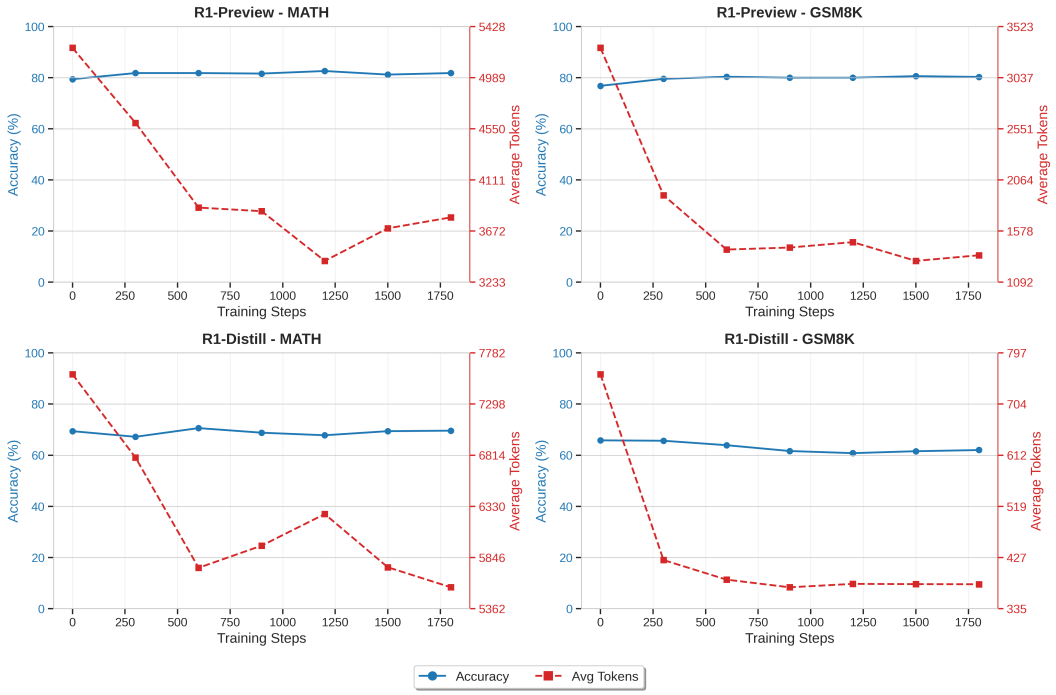

Figure 4: Testing accuracy and average token length when applying the preference optimization algorithm SimPO Meng et al. (2024), trained with preference pairs favoring shorter generations irrespective of correctness.

Motivated by our earlier analysis—which suggests that shorter generations can already achieve strong performance—we investigate whether it is possible to optimize for shorter outputs without relying on correctness signals during training. Specifically, we ask: Can preference optimization that solely prioritizes shorter outputs achieve comparable model performance while reducing the generation length, even without access to ground-truth labels? To explore this, we adopt Simple Preference Optimization (SimPO Meng et al., 2024), an effective offline preference optimization algorithm that uses the average log-probability of a sequence as an implicit reward, with a simplified preference pair construction protocol—we sample just two responses per question and always prefer the shorter one, regardless of correctness. This enables training on entirely unlabeled data while significantly reducing the computational overhead typically associated with large-scale sample generation. We train on the GSM8K and MATH datasets and evaluate performance on their respective test sets. For training data generation, we use a maximum generation length of 8k tokens to accelerate sampling and set both temperature T and top-p to 1. During testing, we allow up to 32k generation length and apply greedy decoding.

Preferring shorter responses reduces generation length while maintaining acceptable accuracy. In Figure 4, we show how accuracy and average token length vary over SimPO training steps. We observe that although accuracy fluctuates during training, it remains within an acceptable range without significant degradation, while the average token length decreases substantially—by approximately 30% to 60%. While the overall generation length is clearly reduced, it remains unclear whether this reduction is primarily due to shortening correct responses, incorrect ones, or both. To investigate this, we separately track the token length for correct and incorrect responses across training steps, as shown in Figure 5.

Length Reduction for Incorrect Responses Contributes More to Length Reduction. Figure 5 reveals that incorrect responses are generally much longer than correct ones. Consequently, the

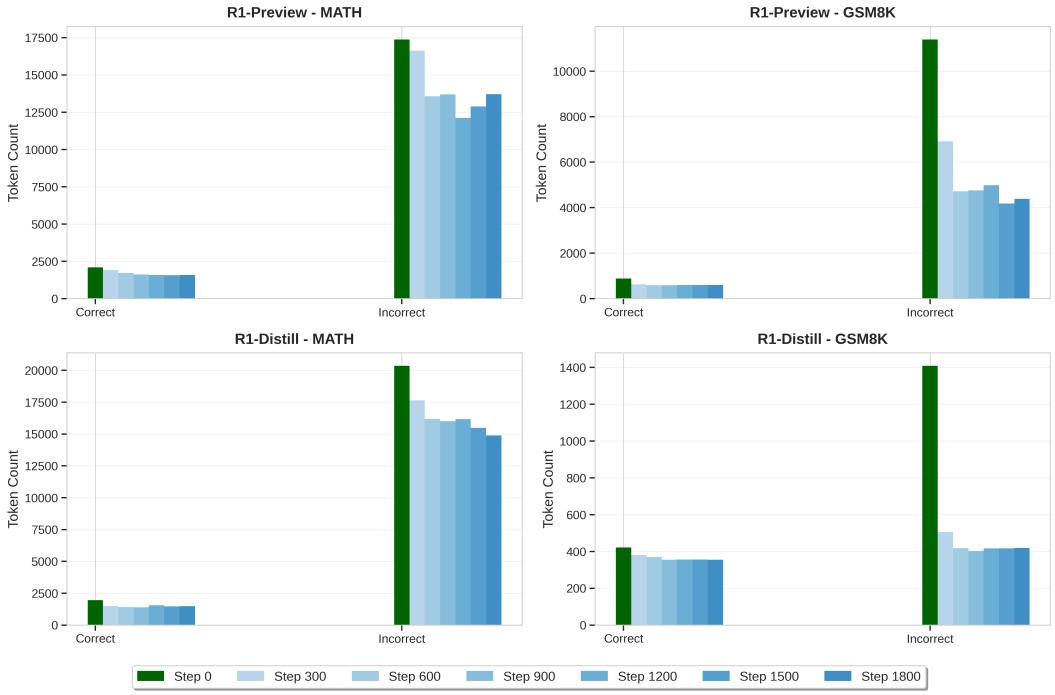

Figure 5: Average token length for correct and incorrect responses across different training steps.

observed reduction in overall generation length is mainly driven by shortening the incorrect responses. On the GSM8K dataset, the average token length of incorrect responses drops significantly during the training, while comparably, the length reduction is less prominent for incorrect responses in the MATH dataset. Additional results, including detailed reduction rates for correct and incorrect responses and trade-offs between accuracy and token length at different training steps, can be found in Appendix E.

7 Conclusion

In this paper, we empirically study the relationship of generation length and final answer correctness from both sample level and question level. We find that current reasoning-oriented LLMs struggle to assess problem difficulty and fail to adjust their generation length appropriately for questions they ultimately answer incorrectly—suggesting a tendency toward underthinking. Conversely, for easier problems that they consistently solve correctly, the models are better at estimating difficulty and adjusting response length. However, they still tend to generate unnecessarily long responses in these cases, indicating a pattern of overthinking. As a result, simply preferring shorter generations—without access to correctness signals—may reduce token length without significantly harming performance.

8 Limitations and Future Work

To ensure scalability and reduce computational overhead, we use $N = 10$ samples per question when analyzing sample-level trends. While this setting aligns with common practice and suffices to reveal consistent patterns, future work could benefit from using larger N to further strengthen statistical reliability and enable finer-grained insights. While our analysis provides new insights into the relationship between generation length and reasoning accuracy, we focus on two widely-used reasoning datasets and two representative reasoning-capable LLM. Extending this analysis to a broader set of models—with different training strategies—and dataset with varying difficulty levels would help assess the generality of our observations and clarify whether these patterns emerge primarily from pretraining or post-training processes such as supervised fine-tuning or reinforcement learning. Thus, we consider it as a future work.